PES University — Dept. of CSE

# Lab 1: Requirements Engineering — Requirements Table

Problem Statement #07: Hostel Maintenance & Issue Ticketing System

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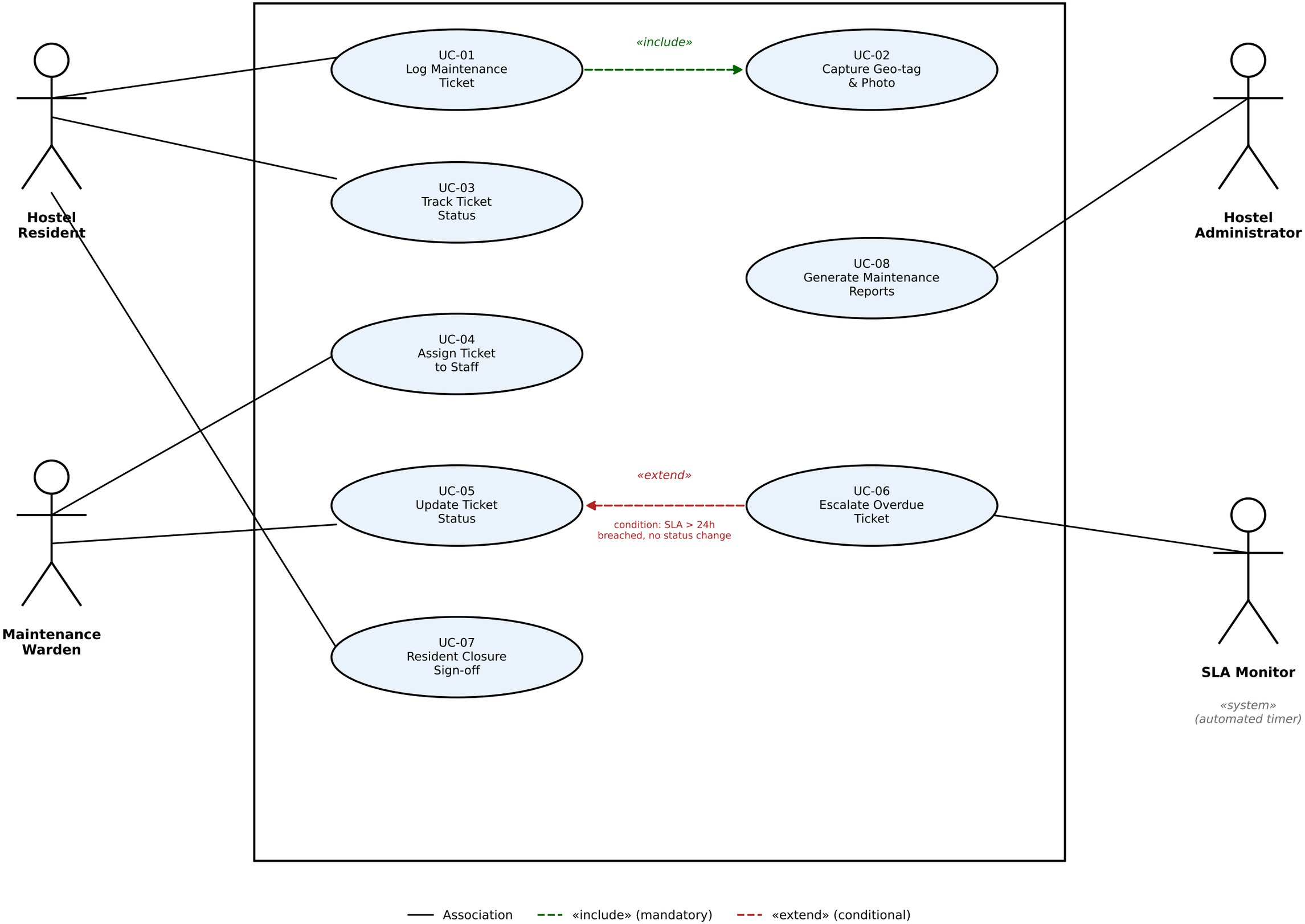
| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| FR-001 | Functional | The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity. | High | Pass: Ticket is created, assigned a unique ID, and routed to warden queue.<br>Fail: Incomplete form accepted without room tag. | Core intake function; without a room tag and severity, tickets cannot be triaged or routed. |
| FR-002 | Functional | The system shall allow a resident to view the real-time status of any ticket they have raised (e.g., Open, Assigned, In Progress, Resolved). | High | Pass: Selecting a ticket ID displays current status and last-updated timestamp within 2 seconds.<br>Fail: Status shown does not match the warden's latest update. | Residents need visibility into progress without contacting the warden directly, reducing follow-up calls. |
| FR-003 | Functional | The system shall allow a maintenance warden to assign an open ticket to a specific maintenance staff member based on issue category (plumbing, electrical, carpentry). | High | Pass: Ticket status changes to 'Assigned' and the selected staff member is notified with ticket details.<br>Fail: Ticket remains unassigned or notification is not sent. | Correct routing to the right skill-set reduces resolution time and avoids misassigned jobs. |
| FR-004 | Functional | The system shall allow the maintenance warden to update a ticket's status (Open / In Progress / Resolved) with a mandatory timestamp and optional remarks. | Medium | Pass: Status change is saved, timestamped, and reflected instantly in the resident's ticket view.<br>Fail: Status updates without a timestamp or fails to sync to the resident's view. | An auditable status trail is needed for SLA tracking and dispute resolution. |
| FR-005 | Functional | The system shall require the resident to confirm closure (sign-off) of a resolved ticket, with an option to reopen if the issue persists within 48 hours. | Medium | Pass: Ticket is marked 'Closed' only after resident confirmation; reopening within 48 hours restores 'In Progress' status.<br>Fail: Ticket is closed automatically without resident confirmation. | Resident sign-off validates that the reported issue was genuinely fixed, preventing premature closures. |
| NFR-001 | Nonfunctional (Performance & Security) | The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change. | High | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. | Guarantees urgent issues (e.g., electrical faults) are not silently ignored past a safe threshold. |
| NFR-002 | Nonfunctional (Security & Reliability) | The system shall encrypt all resident personal data and ticket photo attachments both at rest and in transit, and restrict ticket record access to authenticated wardens and administrators only. | High | Pass: Security audit / penetration test confirms AES-256 (or equivalent) encryption at rest, TLS in transit, and role-based access control blocks unauthorized viewers.<br>Fail: Any ticket record or photo is retrievable by an unauthenticated or unauthorized role. | Tickets contain room numbers, resident identity, and photos of private living spaces, which requires strong data protection and access control. |

Stakeholders / Actors: Hostel Resident, Maintenance Warden, Hostel Administrator, SLA Monitor (automated system).

# Use-Case Diagram: Hostel Maintenance & Issue Ticketing System

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**«System» Hostel Maintenance & Issue Ticketing System**



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# Use-Case Flow Specification

Problem Statement #07: Hostel Maintenance & Issue Ticketing System

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**Use Case:** UC-01 — Log Maintenance Ticket

**Primary Actor:** Hostel Resident
**Supporting Actors:** Maintenance Warden (receives routed ticket), SLA Monitor (starts SLA timer on creation)
**Related Relationships:** «include» UC-02 Capture Geo-tag & Photo (always executes as part of this use case)

## Preconditions

- Resident is authenticated on the hostel ticketing app/portal.
- Resident is assigned to a valid, registered room/block in the system.
- Device location services and camera access are enabled (for geo-tag and photo capture).

## Postconditions

- A new ticket exists with a unique Ticket ID, status 'Open', and is visible in the assigned warden's queue.
- The 24-hour SLA timer for the ticket's priority level has started.
- The resident can track the ticket status from their dashboard.

## Main Success Scenario

1. Resident selects 'Report New Issue' from the home screen.
2. System displays a ticket form requesting: room number, issue category (Plumbing / Electrical / Carpentry), severity (Low/Medium/High), and description.
3. Resident fills in the room number, selects category and severity, and enters a short description.
4. Resident attaches at least one photo of the issue. System invokes UC-02 (Capture Geo-tag & Photo) to auto-tag the resident's current location and timestamp the photo. [«include»]
5. Resident reviews the auto-filled room tag and photo, then taps 'Submit'.
6. System validates that all mandatory fields (room number, category, severity, at least one photo) are present.
7. System generates a unique Ticket ID and sets status to 'Open'.
8. System routes the ticket to the appropriate maintenance warden's queue based on issue category and hostel block.
9. System starts the SLA countdown timer corresponding to the selected severity level.
10. System displays a confirmation screen with the Ticket ID and expected acknowledgement window; the resident may now track it under 'My Tickets'.

## Alternate Flow — 6a. Incomplete Form Submitted

**6a.1.** System detects a missing mandatory field (e.g., no room number tag or no photo attached).

**6a.2.** System rejects the submission and displays an inline error indicating the specific missing field(s).

**6a.3.** Resident corrects the highlighted field(s) and resubmits the form.

**6a.4.** Flow resumes at Main Success Scenario step 6.

***Note:*** *If the resulting ticket later breaches its 24-hour SLA without a status change, UC-06 (Escalate Overdue Ticket) fires automatically as an «extend» of UC-05 (Update Ticket Status) — see NFR-001.*